



WOLLO University

Department of software engineering

Exit multiple choice questions for

Fundamental networking

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1.

What does PoP stand for?

Pre Office Protocol

Post Office Protocol

Protocol of Post

None

Hide

Wrong Answer

Answer - B) PoP stands for Post Office Protocol

2.

What is the port number of PoP?

35

43

110

25

Hide

Wrong Answer

Answer - C) The ports number of PoP is 110

3.

What is the number of layers in the OSI model?

2 layers

4 layers

7 layers

9 layers

Hide

Wrong Answer

Answer - C) OSI model consists of 7 layers.

4.

The full form of OSI is?

Operating System interface

Optical System interconnection

Operating System Internet

Open system interconnection

Hide

Wrong Answer

Answer - D) OSI stands for Open system interconnection.

5.

Identify the layer which provides service to the user.

Session layer

Application layer

Presentation error

Physical layer

Hide

Wrong Answer

Answer - B) Application layer provides service to the user.

6.

What is a HUB?

Software

Computing device

Network device

Calculating device

Hide

Wrong Answer

Answer - C) A HUB is a network device

7.

What does a set of rules define?

SMTP

FTP

IMAP

Protocol

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Hide

Wrong Answer

Answer - Protocol defines a set of rules.

8.

Identify among the following which is mainly used to host web site.

Mail server

Webserver

Database server

None

Hide

Wrong Answer

Answer - B) Web server is mainly used to host web site.

9.

Identify the full form of HTTP?

HyperText Transfer Protocol

HyperText Transfer Package

Hyper Transfer Text Package

Hyper Transfer Text Practice

Hide

Wrong Answer

Answer - A) HTTPS stands for HyperText Transfer Protocol

10.

Identify the protocol primarily used for browsing data.

FTP

TCP

TFTP

HTTP

Hide

Wrong Answer

Answer - D) HTTP is used for browsing data.

11.

Identify the total versions of IP.

1

2

3

4

Hide

Wrong Answer

Answer - B) IP has 2 versions, IPV4 and IPV6.

12.

Identify the first network which was based on TCP/IP protocol.

ARPANET

HUB

Ethernet Card

Router

Hide

Wrong Answer

Answer - A) ARPANET was the first network that was based on TCP/IP protocol.

13.

Choose among the following, which is the most common internet protocol.

PPP

FTP

TCP/IP

SMTP

Hide

Wrong Answer

Answer - D) SMTP is the most commonly used internet protocol.

14.

What does TCP/IP stand for?

Telephone control protocol / Internet Protocol

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Transmission control protocol/Internet protocol

Transmission control protocol/International protocol

None

Hide

Wrong Answer

Answer - B) TCP/IP stands for Transmission control protocol/Internet protocol.

15.

Which of the following layer isn't present in the TCP/IP model but is included in the OSI model?

Network layer

Session layer

Application layer

Transport layer

Hide

Wrong Answer

Answer - B) The TCP/IP model does not contain a session layer.

16.

What is the collection of the hyperlinked document on the internet known as?

HTML

Email

WWW

Internet

Hide

Wrong Answer

Answer - The collection of the hyperlinked document on the internet known as WWW.

17.

What is the location of a resource on the internet given by?

Email

IP

Protocol

URL

Hide

Wrong Answer

Answer - D) The location of a resource on the internet is given by URL

18.

Identify the incorrect network topology,

Bus

Star

P2P

Ring

Hide

Wrong Answer

Answer - C) P2P is not network topology.

19.

Choose the port number of FTP.

23

21

110

143

Hide

Wrong Answer

Answer - B) The port number of FTP is 21.

20.

What is the length of the IPv4 address?

8 bits

16 bits

32 bits

128 bits

Hide

Wrong Answer

Answer - C) The length of the IPv4 address is 32 bits.

21.

What is the length of the IPv6 address?

8 bits

16 bits

32 bits

128 bits

Hide

Wrong Answer

Answer - D) The length of the IPv6 address is 128 bits.

22.

What is the term used when the main server sends mail to another mail server?

FTP

SMTP

TCP/IP

MIME

Hide

Wrong Answer

Answer - B) SMTP is the term used when the main server sends mail to another mail server.

23.

What is the port number of SMTP?

110

143

25

99

Hide

Wrong Answer

Answer - C) The port number of SMTP is 25.

24.

What does MIME stand for?

Multipurpose Internet Mail Extra

Multi Internet Mail End

Multipurpose Internet Mail Email

Multipurpose Internet Mail Extension

Hide

Wrong Answer

Answer - D) MIME stands for Multipurpose Internet Mail Extension.

25.

What does port number 143 refer to?

SMTP

FTP

IMAP

POP

Hide

Wrong Answer

Answer - C) Port number 143 refers to IMAP.

26.

Identify among the following the network device used to connect two dis-similar types of networks.

Switch

Hub

Bridge

Gateway

Hide

Wrong Answer

Answer - D) Gateway is used to connect two dis-similar types of networks.

27.

Identify the device used to boost up a weak signal.

Modem

Repeater

Switch

Router

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Hide

Wrong Answer

Answer - B) Repeater is used to boost up a weak signal.

28.

What does MAC stand for?

Media Access Control

Mass Access Control

Media Access Carriage

None

Hide

Wrong Answer

Answer - A) MAC stands for Media Access Control.

29.

What is the length of the MAC address?

8 bits

16 bits

32 bits

48 bits

Hide

Wrong Answer

Answer - D) Length of MAC address is 48 bits.

30.

Which of the following is used to allocate and manage resources for a network?

Bluetooth

Server

Node

None of the above

Hide

Wrong Answer

Answer - B) Server is used to allocate and manage resources for a network.

31.

The arrangement where all data pass through a central computer is known as

Ring topology

Mesh topology

Star topology

Bus topology

Hide

Wrong Answer

Answer - C) The arrangement where all data pass through a central computer is known as Star topology.

32.

What of the following device is used in the network layer?

Application gateway

Switch

Router

Repeaters

Hide

Wrong Answer

Answer - C) Router is used in the network layer.

33.

Identify if the following statement is True or False: Network Interface Card(NIC) is an I/O device.

True

False

Depends on usage

None

Hide

Wrong Answer

Answer - A) NIC is an I/O device.

34.

What is required to use a Simple Network Management System?

Servers

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Protocols

Rules

IP

Hide

Wrong Answer

Answer - C) Rules are required to use a Simple Network Management System.

35.

Identify the major difference between SNMPv3 and SNMPv2.

Classification

Integration

Management

Enhanced security

Hide

Wrong Answer

Answer - D) The major difference between SNMPv3 and SNMPv2 is Enhanced security.

36.

Identify the network which extends a private network across a public network.

Storage Area network

Virtual private network

Enterprise Private network

Local area network

Hide

Wrong Answer

Answer - B) VPN extends a private network across a public network.

37.

Identify the layer which is responsible for data translating.

Network

Datalink

Presentation

Application

Hide

Wrong Answer

Answer - C) Presentation layer is used for data translating.

38.

Identify the layer which determines the interface of the system with the user.

Network

Datalink

Presentation

Application

Hide

Wrong Answer

Answer - D) Application layer is used to determine the interface of the system with the user.

39.

Which of the following topology arrangements is a point-to-point line configuration?

Ring

Mesh

Star

All of the above

Hide

Wrong Answer

Answer - D) All of the above consists of a point-to-point line configuration.

40.

Identify the device which links two homogeneous packed broadcast local networks.

Hub

Router

Bridge

Gateway

Hide

Wrong Answer

Answer - C) Bridge is used to link two homogeneous packed broadcast local networks.

41.

Why are parity bits used?

To encrypt data

To detect error

To identify user

None

Hide

Wrong Answer

Answer - B) Parity bits are used to detect the error.

42.

Identify among the following which belongs to class A.

121.12.12.248

128.12.12.248

129.12.12.248

130.12.12.248

Hide

Wrong Answer

Answer - A) 121.12.12.248 belongs to class A as the first octet value in the address lies between [0, 127].

43.

What does LAN stand for?

Local Array Network

Local Area Network

Local Area Net

None

Hide

Wrong Answer

Answer - B) LAN stands for Local Area Network.

44.

Who keeps the private key in asymmetric key cryptography?

Sender

Receiver

Both Sender and Receiver

None

Hide

Wrong Answer

Answer - B) Receiver keeps the private key in asymmetric key cryptography.

45.

Calculate the maximum efficiency of pure ALOHA at $G = 0.5$?

16%

1.89%

18.4%

18.999%

Hide

Wrong Answer

Answer - D) The maximum efficiency of pure ALOHA is given by $G * e^{(-2G)}$

Here $G = 0.5$

So, $0.5 * e^{(-2 * 0.5)} = 0.5 * e = 18.4\%$

46.

Identify the switching method in which the message is divided into small packets.

Virtual switching

Packet switching

Message switching

None

Hide

Wrong Answer

Answer - B) In packet switching message is divided into small packets.

47.

What is a proxy server also known as?

Application-level gateway

Proxy tools

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Application proxy

None

Hide

Wrong Answer

Answer - A) Proxy server is also known as Application-level gateway.

48.

Identify among the following servers which allow LAN users to share data.

Communication server

Point server

Data server

File server

Hide

Wrong Answer

Answer - D) File server allows LAN users to share data.

49.

Choose the correct formula for the total vulnerable time value of pure ALOHA.

$\frac{1}{2} T_{fr}$

T_{fr}

$2 * T_{fr}$

$4 * T_{fr}$

Hide

Wrong Answer

Answer - C) The total vulnerable time value of pure ALOHA is $2 * T_{fr}$.

50.

Choose among the following which is a bit-oriented protocol.

SSL

HDLC

HTTP

All of the above

Hide

Wrong Answer

Answer - B) HDLC or High-level link control is a bit-oriented protocol that is used to transmit info from one network to another.

1) Which of these is a standard interface for serial data transmission?

- a. ASCII
- b. RS232C
- c. 2
- d. Centronics

Hide Answer Workspace

Answer: (b) RS232C

Explanation: The RS232C is a standard interface for serial data transmission that defines the protocol and physical interface for transmitting serial data fairly easily between associated appliances and computers.

2) Which type of topology is best suited for large businesses which must carefully control and coordinate the operation of distributed branch outlets?

- a. Ring
- b. Local area
- c. Hierarchical
- d. Star

Hide Answer Workspace

Answer: (d) Star

Explanation: The star topology is the best network topology for large businesses because it is simple to control and coordinate from the central computer.

3) Which of the following transmission directions listed is not a legitimate channel?

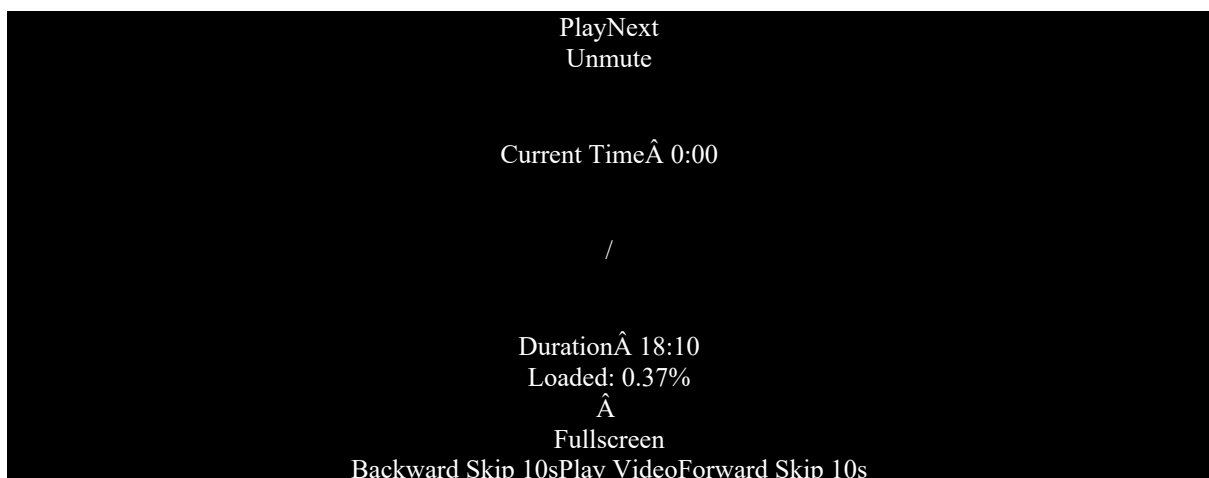
- a. Simplex
- b. Half Duplex
- c. Full Duplex
- d. Double Duplex

Hide Answer Workspace

Answer: (d) Double Duplex

Explanation: Double duplex is not a legitimate channel for transmission in computer network.

4) "Parity bits" are used for which of the following purposes?



- a. Encryption of data
- b. To transmit faster

- c. To detect errors
- d. To identify the user

Hide Answer Workspace

Answer: (c) To detect errors

Explanation: The parity bit is also known as the check bit, and has a value of 0 or 1. It is used for error detection for blocks of data.

5) What kind of transmission medium is most appropriate to carry data in a computer network that is exposed to electrical interferences?

- a. Unshielded twisted pair
- b. Optical fiber
- c. Coaxial cable
- d. Microwave

Hide Answer Workspace

Answer: (b) Optical fiber

Explanation: The optical fiber is made of glass or plastic. In this cable, the transmission of data occurs in the form of light rather than the electric current, so this cable provides higher data transfer speed than other cables.

6) A collection of hyperlinked documents on the internet forms the ??

- a. World Wide Web (WWW)
- b. E-mail system
- c. Mailing list
- d. Hypertext markup language

Hide Answer Workspace

Answer: (a) World Wide Web (WWW)

Explanation: World Wide Web (WWW) creates a collection of hyperlinked documents on the Internet.

7) The location of a resource on the internet is given by its?

- a. Protocol

- b. URL
- c. E-mail address
- d. ICQ

Hide Answer Workspace

Answer: (b) URL

Explanation: A URL (Uniform Resource Locator) is a database connection that describes the database's location on a computer network and the retrieval process. A URL is a different form of URI (Uniform Resource Identifier) although the two words are used interchangeably by many people.

8) The term HTTP stands for?

- a. Hyper terminal tracing program
- b. Hypertext tracing protocol
- c. Hypertext transfer protocol
- d. Hypertext transfer program

Hide Answer Workspace

Answer: (c) Hypertext transfer protocol

Explanation: The term HTTP stands for Hypertext transfer protocol.

9) A proxy server is used as the computer?

- a. with external access
- b. acting as a backup
- c. performing file handling
- d. accessing user permissions

Hide Answer Workspace

Answer: (a) with external access

Explanation: A proxy server is a computer that acts as a gateway between a user's computer and the Internet. The proxy server is also called application level gateway. By this the client computer can establish indirect network connection to another network.

10) Which one of the following would breach the integrity of a system?

- a. Looking the room to prevent theft
- b. Full access rights for all users
- c. Fitting the system with an anti-theft device
- d. Protecting the device against willful or accidental damage

Hide Answer Workspace

Answer: (b) Full access rights for all users

Explanation: None

11) Which software prevents the external access to a system?

- a. Firewall
- b. Gateway
- c. Router
- d. Virus checker

Hide Answer Workspace

Answer: (a) Firewall

Explanation: A firewall is a network securing software that prevents unauthorized users and dangerous elements from accessing the network. Software firewall acts as a filter for our network which prevents harmful information.

12) Which one of the following is a valid email address?

- a. javat@point.com
- b. gmail.com
- c. tpoint@.com
- d. javatpoint@books

Hide Answer Workspace

Answer: (a) javat@point.com

Explanation: A proper email address is a combination of the email prefix and email domain, both in appropriate formats. The domain appears to the right side of the @ symbol and the prefix appears to the left side of the @ symbol.

For example, in this email address javat@point.com, "javat" is the prefix, and "point.com" is the domain.

13) Which of the following best describes uploading information?

- a. Sorting data on a disk drive
- b. Sending information to a host computer
- c. Receiving information from a host computer
- d. Sorting data on a hard drive

Hide Answer Workspace

Answer: (b) Sending information to a host computer

Explanation: None

14) Which one of the following is the most common internet protocol?

- a. HTML
- b. NetBEUI
- c. TCP/IP
- d. IPX/SPX

Hide Answer Workspace

Answer: (c) TCP/IP

Explanation: TCP/IP is the most common internet protocol because it is the most widely used network protocol.

15) Software programs that allow you to legally copy files and give them away at no cost are called which of the following?

- a. Probe ware
- b. Timeshare
- c. Shareware
- d. Public domain

Hide Answer Workspace

Answer: (d) Public domain

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Explanation: Public domain software can be modified, distributed or sold by anyone without any attention, but no one can ever own it.

16) The term FTP stands for?

- a. File transfer program
- b. File transmission protocol
- c. File transfer protocol
- d. File transfer protection

Hide Answer Workspace

Answer: (c) File transfer protocol

Explanation: The term FTP stands for File transfer protocol.

17) At what speed does tele-computed refer?

- a. Interface speed
- b. Cycles per second
- c. Baud rate
- d. Megabyte load

Hide Answer Workspace

Answer: (c) Baud rate

Explanation: In telecommunication, baud rate is a specific unit of the speed rate. It is one of the significant functions that determine the speed of the communication over the data channel.

18) Which one of the following is not a network topology?

- a. Star
- b. Ring
- c. Bus
- d. Peer to Peer

Hide Answer Workspace

Answer: (d) Peer to Peer

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Explanation: Peer to Peer network is a network to which all computers are used the same resources and rights as other computers. Its network designed primarily for the small local area.

19) The maximum length (in bytes) of an IPv4 datagram is?

- a. 32
- b. 1024
- c. 65535
- d. 512

Hide Answer Workspace

Answer: (c) 65535

Explanation: None

20) Which of the following statements could be valid with respect to the ICMP (Internet Control Message Protocol)?

- a. It reports all errors which occur during transmission.
- b. A redirect message is used when a router notices that a packet seems to have been routed wrongly.
- c. It informs routers when an incorrect path has been taken.
- d. The "destination unreachable" type message is used when a router cannot locate the destination.

Hide Answer Workspace

Answer: (b) A redirect message is used when a router notices that a packet seems to have been routed wrongly.

Explanation: None

21) The IP network 192.168.50.0 is to be divided into 10 equal sized subnets. Which of the following subnet masks can be used for the above requirement?

- a. 255.243.240
- b. 255.255.0.0
- c. 255.255.0
- d. 255.255.255

Hide Answer Workspace

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Answer: (c) 255.255.255.0

Explanation: This address belongs to class C, so 8 bits are reserved for the host ID. 24 bits are reserved for network ID.

22) When the mail server sends mail to other mail servers it becomes ____ ?

- a. SMTP client
- b. SMTP server
- c. Peer
- d. Master

Hide Answer Workspace

Answer: (a) SMTP client

Explanation: The SMTP client is an organization that allows sending emails using the SMTP server. SMTP servers can't send emails to other SMTP servers separately. It is based on client-server architecture.

23) The length of an IPv6 address is?

- a. 32 bits
- b. 64 bits
- c. 128 bits
- d. 256 bits

Hide Answer Workspace

Answer: (c) 128 bits

Explanation: An IPv6 address is 128 bits long, which has 2^{128} address space.

24) Consider the following:

- 1. Twisted pair cables
- 2. Microwaves and Satellite Signals
- 3. Repeaters
- 4. Analog Transmissions
- 5. Fiber optics

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Which of the above is consider as (a) signal transmission medium is data communications?

- a. (1) and (5)
- b. (1) and (2)
- c. (1) (2) and (5)
- d. (1) (2) (3) and (5)

Hide Answer Workspace

Answer: (c) (1) (2) and (5)

Explanation: None

25) Which of the following address belongs class A?

- a. 121.12.12.248
- b. 130.12.12.248
- c. 128.12.12.248
- d. 129.12.12.248

Hide Answer Workspace

Answer: (a) 121.12.12.248

Explanation: 121.12.12.248 address belongs the class A because the first octet value of this address lies between 0 and 127.

26) Which of the following is correct IPv4 address?

- a. 124.201.3.1.52
- b. 01.200.128.123
- c. 300.142.210.64
- d. 10110011.32.16.8
- e. 128.64.0.0

Hide Answer Workspace

Answer: (e) 128.64.0.0

Explanation: 128.64.0.0 is correct IPv4 address because IPv4 is a standard numbering system that uses four integers from 0 to 255. The IP address is a group of numbers that identify user system on the network.

27) Which of the following IP addresses can be used as (a) loop-back addresses?

- a. 0.0.0.0
- b. 127.0.0.1
- c. 255.255.255.255
- d. 0.255.255.255

Hide Answer Workspace

Answer: (b) 127.0.0.1

Explanation: A loopback address is a special IP address whose IP address is between 127.0.0.1 to 127.255.255.255. It is reserved for loopback. It doesn't require a physical connection to a network.

28) The term WAN stands for?

- a. Wide Area Net
- b. Wide Access Network
- c. Wide Area Network
- d. Wide Access Net

Hide Answer Workspace

Answer: (c) Wide Area Network

Explanation: The term WAN stands for Wide Area Network.

29) Which of the following cannot be used as a medium for 802.3 ethernet?

- a. A thin coaxial cable
- b. A twisted pair cable
- c. A microwave link
- d. A fiber optical cable

Hide Answer Workspace

Answer: (c) A microwave link

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Explanation: A microwave link cannot be used as a medium for 802.3 ethernet, because a microwave link is a transmission network that utilizes a beam of radio waves in the microwave frequency spectrum to relay video, audio, or data between two places.

30) What IP address class allocates 8 bits for the host identification part?

- a. Class A
- b. Class B
- c. Class C
- d. Class D

Hide Answer Workspace

Answer: (c) Class C

Explanation: In class C only, 8 bits are reserved for the host ID, and 24 bits are reserved for network ID.

31) The term IANA stands for?

- a. Internet Assigned Numbers Authority
- b. Internal Assigned Numbers Authority
- c. Internet Associative Numbers Authoritative
- d. Internal Associative Numbers Authority

Hide Answer Workspace

Answer: (a) Internet Assigned Numbers Authority

Explanation: The term IANA stands for Internet Assigned Numbers Authority.

32) How many versions available of IP?

- a. 6 version
- b. 4 version
- c. 2 version
- d. 1 version

Hide Answer Workspace

Answer: (c) 2 version

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Explanation: There are only two IP versions available in the present: IP version 4 (IPv4) and IP version 6 (IPv6).

33) Which layer of the TCP / IP stack corresponds to the OSI model transport layer?

- a. Host to host
- b. Application
- c. Internet
- d. Network Access

Hide Answer Workspace

Answer: (a) Host to Host

Explanation: The host to host layer conforms the transport layer of the OSI model. This layer is responsible for the final correspondence and error-free distribution of data.

34) An Aloha network uses an 18.2 kbps channel for sending message packets of 100 bits long size. Calculate the maximum throughput.

- a. 5999
- b. 6900
- c. 6027
- d. 5027

Hide Answer Workspace

Answer: (c) 0.6027

Explanation: In Pure Aloha, Efficiency = 18.4%

Usable bandwidth for 18.2 kbps = $18.2 * 0.18 = 3.276$ kbps

Therefore, the maximum throughput of Pure Aloha

$$= 1/2e * 3.276$$

$$= (18.4 * 3.276) / 100$$

$$= \mathbf{0.6027}$$

35) On a simplex data link, which of the following is a possible error recovery technique?

- a. Backward error correction (BEC)
- b. The use of hamming codes
- c. Automatic Repeat Request (ARQ)
- d. Downward error correction (DEC)

Hide Answer Workspace

Answer: (b) The use of hamming codes

Explanation: The hamming codes is an error recovery technique that can be used to detect and correct the errors. It was developed by R.W. Hamming.

36) Which of the statement is correct with regard to Time Division Multiplexing (TDM) and its variants?

- a. Statistical TDM makes efficient use of the bandwidth only if the arrival pattern of the data stream is probabilistic.
- b. TDM requires the transmitter and receiver to be synchronized periodically.
- c. TDM performs efficiently if the arrival pattern of the data stream is probabilistic.
- d. Statistical TDM is efficient if the data stream is deterministic.

Hide Answer Workspace

Answer: (a) and (b)

Explanation: None

37) The term IPv4 stands for?

- a. Internet Protocol Version 4
- b. Internet Programming Version 4
- c. International Programming Version 4
- d. None of these

Hide Answer Workspace

Answer: (a) Internet Protocol Version 4

Explanation: The term IPv4 stands for Internet Protocol Version 4.

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38) The term LAN stands for?

- a. Local Area Net
- b. Local Aera Network
- c. Local Array Network
- d. Local Array Net

Hide Answer Workspace

Answer: (b) Local Area Network

Explanation: The term LAN stands for Local Area Network.

39) Which of the through is share the data of two computer?

- a. Library
- b. Network
- c. Grouping
- d. Integrated system

Hide Answer Workspace

Answer: (b) Network

Explanation: There are many ways to share data between two computers, but a network connection is established before data sharing.

40) In specific, if the systems use separate protocols, which one of the following devices is used to link two systems?

- a. Repeater
- b. Gateway
- c. Bridge
- d. Hub

Hide Answer Workspace

Answer: (b) Gateway

Explanation: If the system used separate protocols, gateway device is used to link two systems.

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41) How many digits of the Data Network Identification Code (DNIC) identify the country?

- a. first three
- b. first four
- c. first five
- d. first six
- e. None of the above

Hide Answer Workspace

Answer: (a) first three

Explanation: The first three digits of the Data Network Identification Code (DNIC) is identify the country (first digit to identify a zone and other two digits to identify the country within the zone).

42) Which of the following methods is used to broadcast two packets on the medium at a time?

- a. Collision
- b. Synchronous
- c. Asynchronous
- d. None of the above

Hide Answer Workspace

Answer: (a) Collision

Explanation: A collision occurs when two or more computers are attempting to transfer data across a network at a time.

43) Which of the following is true with regard to the ping command?

- a. Ping stands for Packet Internet Generator.
- b. The ping command checks the port level connectivity between source destinations end points.
- c. Ping summarizes the packet loss and round-trip delay between two IP end points.
- d. The ping command activates the RARP protocol of the IP layer.

Hide Answer Workspace

Answer: (c) Ping summarizes the packet loss and round-trip delay between two IP end points.

Explanation: None

44) The private key in asymmetric key cryptography is kept by?

- a. Sender
- b. Receiver
- c. Sender and Receiver
- d. None of the these

Hide Answer Workspace

Answer: (b) Receiver

Explanation: Only the receiver keeps the private key. Its purpose is to ensure that the message can only be decrypted by the intended receiver.

45) Which of the following algorithms is not used in asymmetric-key cryptography?

- a. RSA algorithm
- b. Diffie-Hellman algorithm
- c. Electronic code book algorithm
- d. None of the mentioned

Hide Answer Workspace

Answer: (c) Electronic code book algorithm

Explanation: An electronic code book algorithm is a mode of operation for a block cipher, where each frame of text in an encrypted document refers to a data field. In other terms, the same plaintext value would also give the same value for ciphertext.

46) In the cryptography, the sequence of the letters is rearranged by?

- a. Transposition ciphers
- b. Substitution ciphers
- c. Both a and b
- d. None of these

Hide Answer Workspace

Answer: (a) Transposition ciphers

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Explanation: A predefined technique shuffles the sequence of letters in a plaintext message in the transposition ciphers.

47) What is the maximum efficiency of pure aloha at $G = 1/2$?

- a. 1.89
- b. 17.99
- c. 18.999
- d. 18.4

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Answer: (d) 18.4

Explanation: The maximum efficiency of pure aloha formula is $G * e^{-2G}$.

Given, $G = 1/2$

$$= 1/2 \times e^{-2 \times 1/2}$$

$$= 1 / 2e$$

$$= 0.184$$

$$= 18.4\%$$

48) What is the maximum efficiency of slotted aloha at $G = 1$?

- a. 36.8
- b. 35.8
- c. 35.5
- d. 37.8

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Answer: (a) 36.8

Explanation: The maximum efficiency of slotted aloha formula is $G * e^{-G}$.

Given, $G = 1$

$$= 1 \times e^{-1}$$

$$= 1 / e$$

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= 0.368

= 36.8%

49) Which of the following statement is true about error detection techniques used on communications link?

- a. Cyclic Redundancy Check (CRC) sequence can detect as well as correct errors.
- b. Error detection cannot be used on simplex links.
- c. Hamming code can detect up to 3-bit errors.
- d. All of the these

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Answer: (d) All of the these

Explanation: None

50) The correct order of corresponding OSI layers for having functionalities of routing and reconciling machine representation differences with shared access resolution and ASCII test protocol is?

- a. Network, Physical, Transport, Data link
- b. Network, Physical, Data link, Application
- c. Network, Presentation, Data link, Application
- d. Network, Presentation, Physical, Transport

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Answer: (c) Network, Presentation, Data link, Application

Explanation: None